SE LAB 1

PROBLEM 33: **Micro-Lending & Peer Credit Risk Assessor**

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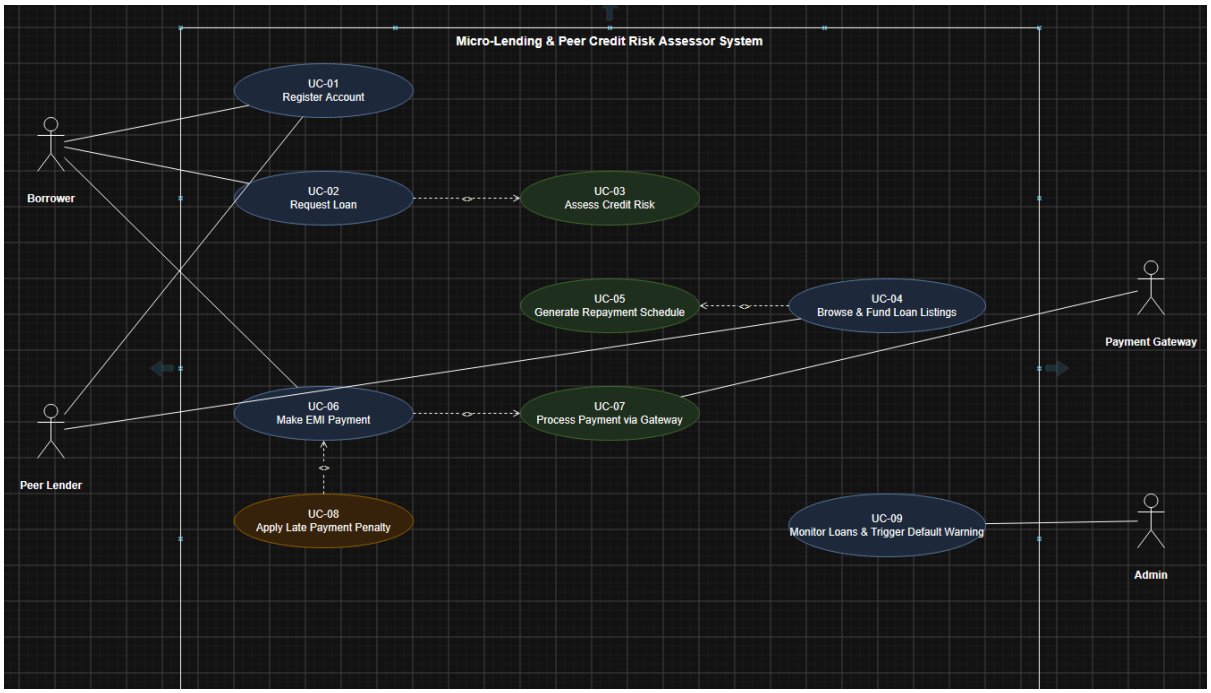
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SECTION: F

1 – Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall compute borrower credit risk ratings based on debt-to-income ratio, repayment history, and employment stability indicators.	High	Pass: Risk tier (Low/Med/High) accurately assigned. Fail: Incomplete financial profile generates credit approval.	Core risk-scoring function of the platform; prevents unqualified borrowers from being approved.	
FR-002	Functional	The system shall allow a borrower to submit a loan request specifying amount, purpose, and repayment tenure.	High	When a borrower submits the form with all required fields, the request appears in "Pending" status with a unique loan request ID within 2 seconds.	Entry point for the lending workflow; without it no risk assessment or funding can occur.	
FR-003	Functional	The system shall generate a structured repayment schedule (EMI amount, due dates, outstanding balance) automatically once a loan request is funded.	High	Pass: Schedule shows correct number of EMIs, each due date exactly 30 days apart, and total of EMIs equals principal + interest. Fail: Schedule is missing or EMI sum does not reconcile with loan amount.	Borrowers and lenders both need a clear, auditable repayment plan to track obligations.	
FR-004	Functional	The system shall allow a peer lender to browse open loan requests filtered by risk tier and fund a selected loan, partially or in full.	Medium	When a lender selects a loan and confirms a funding amount ≤ remaining requested amount, the loan's funded percentage updates immediately and the lender is added to its funder list.	Enables the peer-to-peer matching that is central to the platform's business model.	
FR-005	Functional	The system shall automatically trigger a default warning notification to the borrower and lender(s) when a scheduled EMI payment is not received within 3 days of its due date.	High	Pass: Warning notification is logged and sent within 24 hours of the 3-day grace period expiring. Fail: No warning generated for an overdue EMI past the grace period.	Early default detection protects lender capital and gives borrowers a chance to remediate before penalties escalate.	
NFR-001	Nonfunc. (Performance & Security)	Loan repayment transactions and EMI distribution must execute inside database ACID transactions to prevent double-spending.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Financial correctness and integrity are non-negotiable for a lending platform.	
NFR-002	Nonfunc. (Security)	The system shall encrypt all borrower and lender personal and financial data at rest and in transit using industry-standard encryption (e.g., AES-256, TLS 1.2+).	High	Pass: Security audit/scan confirms all stored PII fields are encrypted and all API traffic uses TLS 1.2 or higher. Fail: Any financial or personal field	Protects sensitive financial/PII data and supports regulatory compliance for a finance platform.	

2 – UML Diagram:



3 – Use Flow Case

Use-Case Flow Specification

Micro-Lending & Peer Credit Risk Assessor — Problem Statement #33

UC-06: Make EMI Payment

Actors: Borrower (primary), Payment Gateway (secondary)

Preconditions

1. Borrower is registered and logged in.
2. Loan has been funded (UC-04 completed) and a repayment schedule exists (UC-03 / FR-003 generated).
3. A due EMI installment exists for the current billing cycle.

Postconditions

1. On success: EMI is marked as paid, outstanding loan balance is updated, and a receipt is generated.
2. On failure: EMI remains marked “due” or “overdue,” and the borrower is notified.

Main Success Scenario

1. Borrower selects “Pay EMI” for an active loan from their dashboard.
2. System displays the due EMI amount, due date, and outstanding balance.
3. Borrower selects a payment method (card / bank transfer / wallet).
4. System sends the payment request to the Payment Gateway («include» UC-07: Process Payment via Gateway).
5. Payment Gateway authorizes and confirms the transaction.
6. System updates the repayment schedule, marking the EMI as “Paid.”
7. System recalculates and updates the outstanding loan balance.
8. System generates and displays/emails a payment receipt to the borrower.
9. Use case ends successfully.

Alternate Flow — 5a. Payment Declined

a) Payment Gateway returns a “declined” status.

b) System displays a “Payment Failed” message and prompts the borrower to retry with the same or a different payment method.

c) If payment fails again after 2 attempts, the system marks the EMI as “Overdue,” logs the failed attempt, and schedules it for the default-warning check («extend» UC-08: Apply Late Payment Penalty, and eventually FR-005’s default warning if unresolved).

d) Use case ends without payment recorded.